

CSC360 - Computer Graphics and Digital Image Processing

27/8/26 Reflection

Topic: JavaFX, Maven, CI/CD and Testing

Today in class, we talked about how a JavaFX project is set up, executed, tested, and how it gets packaged. We also covered how dependencies get managed in Maven and how a project can be moved from development into deployment.

JavaFX and Maven

We talked about the procedure for executing a JavaFX project in IntelliJ IDEA and running it using Maven commands.

- JavaFX needs proper configuration for modules and dependencies.
- Maven takes care of the libraries needed by the project in its repositories.
- target folder stores the compiled files.
- The package step can result in creation of a JAR file.
- JAR is like a zip file and basically comprises the compiled .class files and resources in the project.
- External dependencies won't automatically come in unless the project is built to produce a bundled JAR.

CI/CD

Also, there was one more aspect covered by this course, such as CI/CD. It means how the changes made by the developer are getting implemented into the application.

For example:

- Developer makes and commits the change
- Server receives the latest code
- Integration → changes are merged, built, and tested.
- Deployment → application gets deployed to the remote server.

That helped me to understand the difference between integration of the code and deployment of it.

Java Configuration and Compatibility

Also, there was another issue that we have covered in this course, such as UTF-8 encoding, character encoding, and Java source, target, and release versions.

- UTF-8 specifies the way characters get encoded.
- Different configurations may affect time, speed, memory, and costs.
- Source specifies which language constructs may be used in the code.
- Target specifies the level of compatibility of the produced bytecode.
- Release enables control of both Java language and API compatibility.

Testing and JavaFX Features

Lastly, we were introduced to JUnit and various ways of testing.

- Unit testing → tests parts of the program individually.
- GUI testing → verifies GUI and how it behaves in terms of user interaction.
- Headless testing → tests the program without having to display its GUI.
- JavaFX has controls like buttons and trees.
- FXML gives a way to design JavaFX user interface through declarations.
- JavaFX also provides features regarding web and media.

Key Takeaway

Through today's lecture, I realized that building a JavaFX application requires not only coding but also using other tools to make the application ready for deployment.